

SHORT COMMUNICATION

Modulatory effect of Piperine on mitochondrial antioxidant system in Benzo(a)pyrene-induced experimental lung carcinogenesis

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Summary

Chemoprevention has emerged as a very effective preventive measure against carcinogenesis. Many bioactive compounds present in edible as well in herbal plants have revealed their cancer chemopreventive potential. In the present study, our goal was to investigate the impact of piperine, a principle ingredient of pepper, on alterations of mitochondrial antioxidant system and lipid peroxidation in Benzo(a)pyrene (B(a)P) induced experimental lung carcinogenesis. Oral supplementation of piperine (50 mg/kg body weight) effectively suppressed lung carcinogenesis in B(a)p induced mice as revealed by the decrease in the extent of mitochondrial lipid peroxidation and concomitant increase in the activities of enzymatic antioxidants (superoxide dismutase, catalase and glutathione peroxidase) and non enzymatic antioxidant (reduced glutathione, vitamin E and vitamin C) levels when compared to lung carcinogenesis bearing animals. Our data suggests that piperine may extent its chemopreventive effect by modulating lipid peroxidation and augmenting antioxidant defense system.

Key words: Piperine, mitochondrial antioxidants, lung carcinogenesis, Benzo(a)Pyrene

■ Introduction

Medicinal plants and their active principles have attracted the focus of recent attention as potential chemopreventive agents. Black pepper (*piper nigrum*) and Long pepper (*piper longum*) are used worldwide as spices and as ingredients in traditional system of medicinal plants (Yin Quan, 1983).

Piperine is a major plant alkaloid present in pepper, which is one among the most common spices consumed by a large number of people worldwide. This

compound is known to possess several pharmacological and biochemical actions such as anti microbial, anti inflammatory, hepatoprotective, antimutagenic, etc (Nakini et al. 1998; Dhuley et al. 1993; Rashmeet and Jaswant, 1991). Earlier reports show that piperine enhances the bioavailability of drugs and inhibits xenobiotic (drug) metabolizing enzymes in pulmonary and hepatic tissues (Atal et al. 1985). It also strongly inhibits the constitutive, B(a)p inducible aryl hydrocar-

bon hydroxylase and epoxide hydroxylase activities in rat hepatic tissue (Balakoul and Aruna, 1993; Sing and Rao, 1993). Piperine exhibits antioxidant action in experimental conditions both *in vivo* as well as *in vitro* through its radical quenching effect and by preventing GSH depletion (Mittal and Gupta, 2000). It also inhibits B(a)P activation by lowering B(a)P oxide production and thereby decreases B(a)P cytotoxicity in *in vitro* (Balakoul and Aruna, 1993; Ychu et al. 1994).

The present study was designed to investigate the cytoprotective effect of piperine in an experimental lung carcinogenesis model using mitochondrial lipid peroxidation, enzymatic and nonenzymatic antioxidants as biochemical endpoints of chemoprevention.

■ Materials and Methods

Chemicals

Benzo(a)pyrene was gifted by National Cancer Institute, MRI Missouri, USA. Reduced GSH, DNPH, Bathophenanthroline and piperine were purchased from Sigma chemical company, USA. All other chemicals used were of analytical grade.

Animals

Healthy male Swiss albino mice (17–20 grams) were used throughout the study. They were maintained in a controlled environment condition of temperature and humidity on alternatively 12h light/dark cycles. All animals were fed standard pellet diet (Gold Mohor rat feed, Ms.Hindustan Lever Ltd., Mumbai) and water *ad libitum*.

Experimental protocol

The animals were divided into five groups as depicted in Fig. 1. Group I served as control animals and were given corn oil orally. The Group II animals were treated with Benzo(a)Pyrene ((B(a)P) 50 mg/kg body weight dissolved in corn oil orally) weekly twice for 4 weeks to induce lung carcinogenesis. The Group III animals were treated with piperine (50 mg/kg body weight dissolved in corn oil orally) on alternative days for 16 weeks after they were treated with first dose of B(a)P (as in Group II). The Group IV animals were post initiation treated with piperine (as in G-III) for 10 weeks after the induction of B(a)p(as in group II). The Group V animals were treated with piperine (as in G-III) alone for 16 weeks to study the cytotoxicity (if any) induced by piperine. The initiation and post initiation treatment of piperine were used to study the chemopreventive and/or chemotherapeutic efficacies of piperine in the experimental animals.

Biochemical analysis

Mitochondrial isolation. At the end of the experimental period, the animals were killed by cervical decapitation. Lung and liver were immediately excised, weighed and then analyses used for the isolation of mitochondria. The mitochondria were isolated by the method of Mustafa et al. (1977) and Johnson and Lardy (1967). A 10% homogenate was prepared in 0.25 M sucrose and centrifuged at $600 \times g$ for 10 mins. The supernatant fraction was decanted and centrifuged at $15,000 \times g$ for 10 mins. The resultant mitochondrial pellet was then washed and resuspended in 0.25 M sucrose. The purity of the mitochondria were assessed by assaying the activity of succinate dehydrogenase (SDH) and used for further analyses. Total protein was estimated by the method of Lowry et al. (1951). Super-oxide dismutase (SOD) was estimated by method of Marklund and Marklund (1974), catalase (CAT) by that of Sinha (1972) and glutathione (GPx) by that Rotruck et al. (1972). Reduced glutathione (GSH) was determined by the method of Moron et al. (1979), vitamin E (vit. E) was estimated by the method of Desai (1984) and vitamin C (vit. C) was measured by the method of Omaye et al. (1971). Lipid peroxidation was estimated by the method of Ohkawa et al. (1979).

Statistical analysis

For statistical analysis, one-way analysis of variance (ANOVA) was used, followed by the Newman-Keuls multiple comparison test.

■ Results

Table 1 shows the levels of mitochondrial lipid peroxidation, cellular antioxidant enzymes such as SOD, CAT and GPX and nonenzymatic antioxidants (GSH, vit. E and vit. C) in lung mitochondria of the various experimental groups. Highly significant reduction in the activity of scavenging mitochondria enzymes and non-enzymatic antioxidants, with an increase in the extent of lipid peroxidation in the carcinogenesis bearing mice (Group-II) was observed. These adverse changes were reversed to near normal values in piperine during on and post initiation treated Group-III and Group-IV animals (Lpo, 24.5% reduced, SOD CAT and Gpx 50%, GSH, Vit C and Vit E 75% increased after supplementation of piperine) ($p < 0.05$). However, the piperine alone treated animals (G-V) did not show any significant changes when compared with control animals (G-I).

Table 2 shows increased level of mitochondrial lipid peroxidation ($p < 0.05$) and decreased activities of antioxidant enzymes and levels of non-enzymatic antioxidants ($p < 0.05$) in liver mitochondria of carcinogenesis bearing mice (G-II). These adverse changes were re-

Table 1. Effect of piperine on lipid peroxidation, enzymic and non-enzymic antioxidants in lung mitochondria of control and experimental mice.

Parameters	G-I (Control animals)	G-II (Lung carcinogenesis)	G-III (Piperine treated during initiation)	G-IV (Piperine treated post initiation)	G-V (Piperine treated alone)
LPO	0.23 ± 0.01	0.33 ± 0.03 ^{a*}	0.25 ± 0.02 ^{b*}	0.28 ± 0.02 ^{bNScNS}	0.20 ± 0.02
SOD	2.18 ± 0.24	1.35 ± 0.27 ^{a*}	2.04 ± 0.15 ^{b*}	1.90 ± 0.26 ^{b*cNS}	2.20 ± 0.18
CAT	174 ± 18.0	110 ± 15.0 ^{a*}	165 ± 17.0 ^{b*}	145 ± 12.0 ^{b*c*}	176 ± 18.0
GPX	4.79 ± 0.53	3.13 ± 0.47 ^{a*}	4.70 ± 0.52 ^{b*}	4.10 ± 0.24 ^{b*c*}	4.76 ± 0.51
Reduced GSH	0.77 ± 0.05	0.36 ± 0.04 ^{a*}	0.72 ± 0.08 ^{b*}	0.63 ± 0.05 ^{b*cNS}	0.76 ± 0.06
Vit C	0.39 ± 0.04	0.20 ± 0.03 ^{a*}	0.34 ± 0.04 ^{b*}	0.29 ± 0.03 ^{b*cNS}	0.38 ± 0.04
Vit E	0.45 ± 0.06	0.21 ± 0.03 ^{a*}	0.41 ± 0.03 ^{b*}	0.32 ± 0.04 ^{b*cNS}	0.47 ± 0.03

Each value is expressed as mean ± SD for six rats in each group.

LPO – nmoles of MDA released/mg protein; SOD – units/min/mg protein; CAT – μmoles of H₂O₂ consumed/min/mg protein; GPX – μmoles of GSH oxidized/min/mg protein; protein; Reduced GSH – μg/mg protein; α-tocopherol – μg/mg protein; ascorbic acid – μg/mg protein.

a – As compared with G-I (control animals)

b – As compared with G-II (lung carcinogenesis animals)

c – As compared with G-III (piperine initiation treated animals)

Statistical significance * p < 0.05 and NS – Not significant.

Table 2. Effect of piperine on lipid peroxidation, enzymic and non-enzymic antioxidants in liver mitochondria of control and experimental mice.

Parameters	G-I (Control animals)	G-II (Lung carcinogenesis)	G-III (Piperine treated during initiation)	G-IV (Piperine treated post initiation)	G-V (Piperine treated alone)
LPO	0.25 ± 0.01	0.39 ± 0.03 ^{a*}	0.27 ± 0.02 ^{b*}	0.33 ± 0.03 ^{b*c*}	0.13 ± 0.02
SOD	2.48 ± 0.24	1.59 ± 0.31 ^{a*}	2.42 ± 0.36 ^{b*}	2.13 ± 0.29 ^{b*c*}	2.45 ± 0.31
CAT	194 ± 18.0	126 ± 13.0 ^{a*}	175 ± 11.0 ^{b*}	163 ± 17.0 ^{b*c*}	197 ± 17.0
GPX	5.79 ± 0.53	4.19 ± 0.51 ^{a*}	5.70 ± 0.41 ^{b*}	5.42 ± 0.63 ^{b*c*}	5.77 ± 0.63
Reduced GSH	1.04 ± 0.20	0.67 ± 0.05 ^{a*}	0.96 ± 0.08 ^{b*}	0.84 ± 0.06 ^{b*c*}	1.07 ± 0.23
Vit C	0.46 ± 0.05	0.29 ± 0.04 ^{a*}	0.42 ± 0.03 ^{b*}	0.38 ± 0.04 ^{b*cNS}	0.47 ± 0.03
Vit E	0.57 ± 0.06	0.34 ± 0.04 ^{a*}	0.53 ± 0.06 ^{b*}	0.48 ± 0.05 ^{b*cNS}	0.58 ± 0.04

Each value is expressed as mean ± SD for six rats in each group.

LPO – nmoles of MDA released/mg protein; SOD – units/min/mg protein; CAT – μmoles of H₂O₂ consumed/min/mg protein; GPX – μmoles of GSH oxidized/min/mg protein; protein; Reduced GSH – μg/mg protein; α-tocopherol – μg/mg protein; ascorbic acid – μg/mg protein.

a – As compared with G-I (control animals)

b – As compared with G-II (lung carcinogenesis animals)

c – As compared with G-III (piperine initiation treated animals)

Statistical significance * p < 0.05, NS – Not Significant.

versed to near normal values in piperine during and post initiation treated Group-III and Group-IV animals (Lpo 30% reduced, SOD 52% CAT and GPx 37%, GSH and Vit C 45% and Vit E 55% Increased after supplementation of piperine) (p < 0.05). However, the piperine alone treated animals (G-V) did not show any significant changes when compared with control animals (G-I).

Discussion

Chemoprevention offers a novel approach to control the incidence of lung cancer. A recent international report concluded that high dietary intake of alkaloids and flavonoids reduce further risk of developing lung cancer. The main mechanism responsible for this reduced

risk is the strong antioxidant effect of these substances (Block et al. 1992). Piperine is one such alkaloid and dietary nutrient, which is known to protect against oxidative damage by inhibiting or quenching free radicals and Reactive Oxygen Species (ROS) (Annu et al. 1998).

Mitochondria are the energy reservoir of the cell and the damage inflicted in mitochondria would ultimately result in the reduction of energy production and thereby leads to cell death (Sohal and Dubey, 1994). Sub cellular membranes associated with thiol bearing enzymes represent sensitive sites for B(a)P, causing perturbation of cellular function (Kyu and Byung, 1997). Reactive oxygen species can themselves reduce the activities of antioxidant defense system. In the present study, piperine enhanced mitochondrial enzymatic and nonenzymatic antioxidants and suppressed lipid peroxidation in during and post initiation supplemented animals.

Free radicals react with lipids causing peroxidation, resulting in the release of products such as malondialdehyde, hydro peroxides and hydroxyl radicals. An increase in LPO indicates serious damage to cell membranes, inhibition of several enzymes and cellular function (Thirunavukarasu et al. 2001). Earlier studies *in vitro* show, that MDA can alter protein, DNA, RNA and other biomolecules (Panandiker et al. 1994). In the present study, an increase in the levels of mitochondrial LPO was found during lung carcinogenesis and these levels were significantly reduced after the supplementation of piperine. Piperine has the capacity to scavenge free radicals directly or interfering with the generation of free radicals (Reddy and Lokesh, 1992; Dhuley et al. 1993). Thus, the inhibitory effect of piperine on oxidative damages during and post initiation stages may be attributed to the suppression of B(a)p induced lipid peroxidation.

Antioxidant enzymes are main scavengers of free radicals and the activities of these enzymes are altered during carcinogenesis or after tumour formation (Sun, 1990). The mechanism and implication of this abnormality, however is still unclear. It is well known that SOD, CAT and GPX play an important role as protective enzymes against free radical formation in tissues (Oberly and Buettner, 1974). Several investigators reported that the reduced activities of SOD, CAT and GPX in cancer bearing rats and mice is due to down regulation of the SOD and CAT gene induced by free radicals and also by certain humoral factors (Andersen et al. 1994; Slaga, 1995). The present study indicates a reduction in the activities of mitochondrial SOD, CAT and GPX in lung cancer bearing animals (G-II). On piperine treatment, the activities of these enzymes decreased in G-III and G-IV animals. This may be due to the direct reaction of piperine with superoxide, hydroxyl radical and alkoxyl radicals, which in turn reduces free radical formation and oxidative damages during

lung carcinogenesis. This suggests that the antioxidative and free radical scavenging action of piperine may be a possible mechanism of piperine in protecting against B(a)p induced lung carcinogenesis.

Reduced glutathione, which is a substrate for glutathione peroxidase, neutralizes hydroxyl radicals and singlet oxygen. Since it is present in high concentration in the cells, it protects cells from free radical attack (Knotsky, 1990; Gopalakrishnan et al. 1996). In our present study, the decreased level of glutathione in carcinogenesis-induced animals (G-II) may have been due to excess utilization of this antioxidant by tumor cells for cell proliferation. The reduced GSH in tissues keeps up the cellular levels of vitamin C and vitamin E in active form. These vitamins also exist in interconvertible forms and participate in neutralizing free radicals. When there is reduction in the levels of GSH, cellular levels of ascorbic acid and tocopherol are also lowered (Kagen et al. 1990). This may be the reason for the observed decrease in vitamin C and vitamin E in carcinogenesis bearing mice. But these conditions were found to be reversed in piperine administrated animals in G-III and G-IV. Supplementation of piperine to the experimental animals would have enhanced various cellular antioxidants and thiol contents in tissues, which in turn reduces free radical formation during lung carcinogenesis. It was found in the present study that piperine was more effective in initiation treated group than the post initiation treated with respect to prevention of lung cancer. This may be due to the inhibitory action of piperine on the initiation of B(a)p activation/detoxification process.

In conclusion, to the best of our knowledge, this is the first report showing the chemopreventive and therapeutic efficacy of piperine on lung carcinogenesis and its modulation on oxidative alterations of mitochondrial lipid peroxidation, enzymic and non-enzymic antioxidants. Clearly our study proves the protective role of piperine in decreasing lipid peroxidation and by normalizing the antioxidant defense system.

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